

Franklin Ramirez

 fgramire@uwaterloo.ca |  [LinkedIn](#) |  [Github](#) |  [Personal Portfolio](#)

Education

University of Waterloo

Candidate for Bachelor of Computer Science - Artificial Intelligence Specialization | 3.6 GPA

Present - 2026

Waterloo, ON

Experience

Software Developer

April 2024 - September 2024

Municipal Property Assessment Corporation

Pickering, ON

- Deployed a Retrieval-Augmented Generation (**RAG**) chatbot, leveraging **AWS SageMaker** models, connected to an **OpenSearch** index to deliver contextually accurate responses, enabling IT to resolve issues **35%** faster.
- Converted user queries to embeddings and performed **k-NN searches** against the knowledge corpus index in OpenSearch for dynamic **real-time** model **training**.
- Designed **data pipelines** with **AWS Glue**, **AWS Textract** and **S3**, automating the text extraction and using the NLTK library to prepare data for **embedding generation** and **indexing** in OpenSearch.
- Used **API Gateway** to route requests to **AWS Lambda** for seamless interactions between the bot and AWS.

Lead Machine Learning Engineer

July 2024 - Present

Waterloo Artificial Intelligence Organization

- Leading development of **predictive models** using time-series analysis (**ARIMA/SARIMA**) and supervised learning to **forecast TTC bus delays** based on a synthesis of historical delay, real-time GTFS, and weather data.
- Developing an **ensemble framework** that **aggregates predictions** from various machine learning models, such as SARIMA, Random Forest, Gradient Boosting, to provide a more **robust forecast** of bus delays.
- Designing an **online learning** framework for continuous model improvement using the **river** library and Apache Kafka for data streaming, with the goal of **real-time** adaptation and **improving prediction accuracy**.

Frontend Developer

August 2023 - May 2024

Government Services Integration Cluster - Ministry of Public & Business Service Delivery

Remote

- Implemented the **Jamstack architecture** on web applications for bilingual support, advanced search functionality, and enhanced breadcrumb user navigation to support **millions of Ontario website users** per day.
- Completed **search functionality** and **load capacity** testing to enhance site reliability and user experience.

Projects

StatCan Data Extractor  | *Python, Pandas, SQL, Poetry, Docker, DuckDB, Cloudflare R2, AWS*

- Engineered a comprehensive **data extraction and transformation pipeline** to manage and process large datasets, and convert to parquet (**500GB+**) for storage optimization and schema information to be stored.
- Built SQL scripts in DuckDB with the StatCan REST API for **automated dataset updates**, significantly reducing code complexity and accelerating data retrieval by an estimated **30%**.
- Integrated Cloudflare R2 with DuckDB, utilizing SQL for efficient retrieval and management of StatCan datasets.

DermDetect  | *Vertex AI, PaLM 2, Streamlit, GCP*

- Developed a **skin disease classification tool** with an **86% accuracy** trained on over **15,000** skin images.
- Built in **feature scaling and data normalization**, reducing pre-processing and model training time **by 35%**.

Borealis AI - Stroke Calculator  | *Python, Pandas, Dash, NumPy, TensorFlow, PyTorch*

- Created and trained a binary classification model for stroke-prediction using a random-forest classification algorithm that had a **93% accuracy**, based on analysis of over **430,000 data points**.
- Performed data visualization, such as heat maps, and created a **distribution-based clustering algorithm** to identify variables with the highest **correlation to stroke risk**.

Technical Skills

Languages: Python, Java, C/C++, SQL, Tailwind, JavaScript, TypeScript, R, Scala

Frameworks: React.js, Angular, Node.js, Express.js, Linux, Vue, TensorFlow, PyTorch, NumPy, Scikit-learn, Keras, OpenCV, Docker

Tools: Visual Studio Code, RESTful APIs, Jenkins, MySQL, SQL Server Management Studio, Spark, Jupyter, Figma, AWS Sagemaker, OpenSearchService, AWS CloudFormation, Terraform